

Panel 1 — The Hypothesis

- **Standard view:** QEC assumes noise accumulates over a continuous topology; logical fidelity improves smoothly as redundancy is added.
- **Alternative to test:** error propagation follows a discrete, hierarchical geometry — a Bruhat–Tits tree T_p — and redundancy quantizes in units of $(p+1)$.
- **Signature to look for:** a staircase — flat plateaus followed by sudden jumps — instead of a smooth curve.

STATUS: HYPOTHESIS ONLY — NO EXPERIMENTAL RESULT CLAIMED

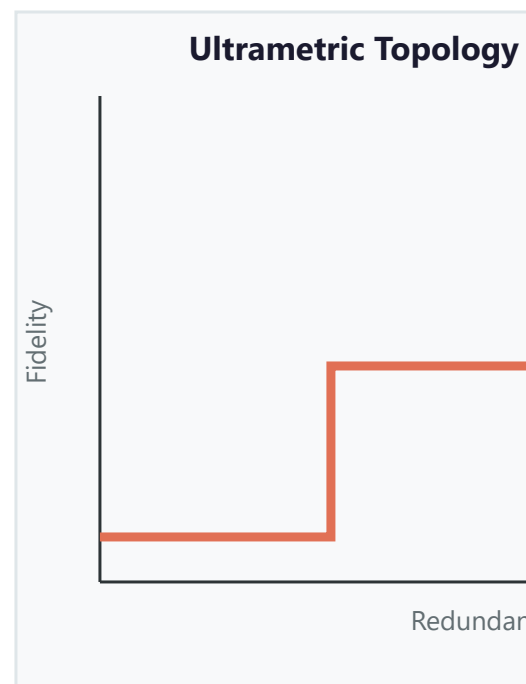
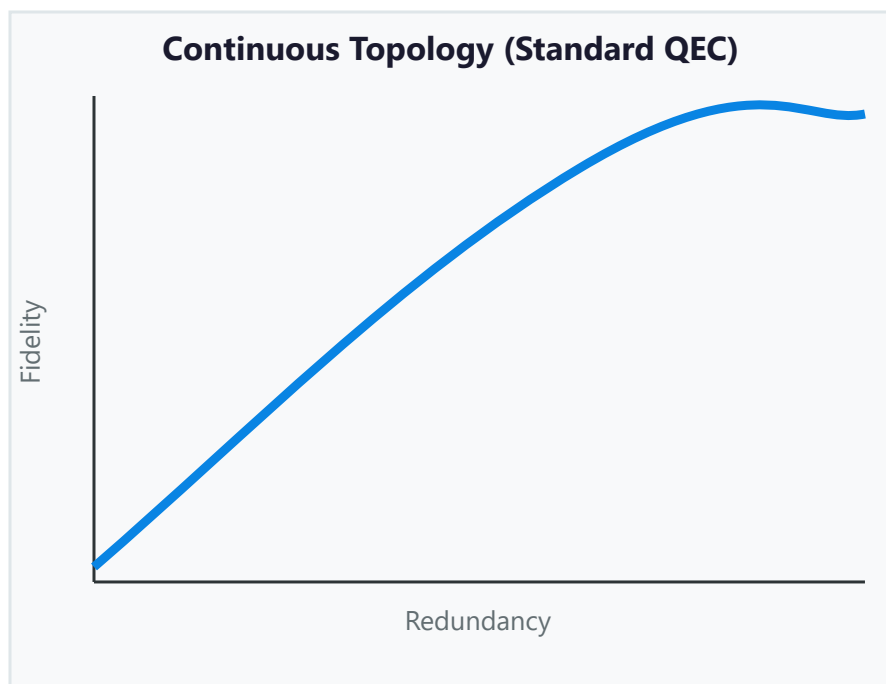


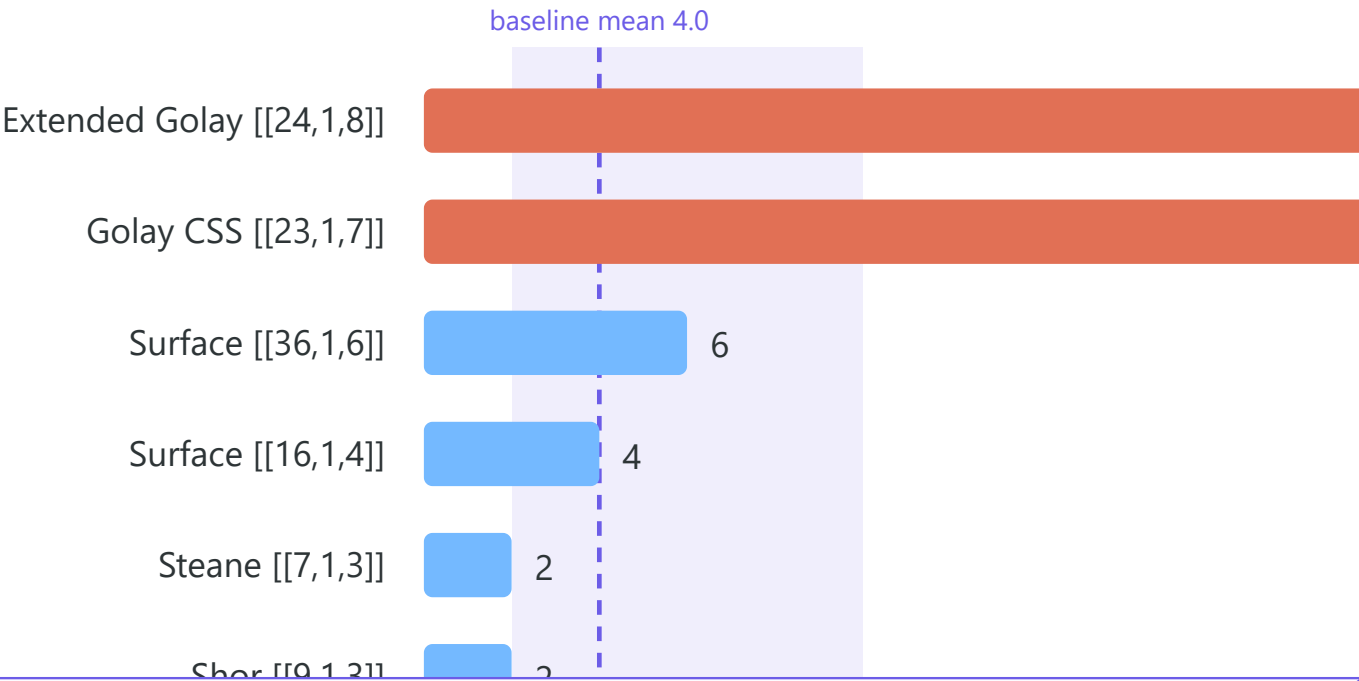
Figure 1: smooth fidelity curve (continuous topology, left) vs. staircase (ultrametric hypothesis, right). Hypothesis only; not observed.

Panel 3 — The Cross-Check I Already Ran
(it mostly failed — honestly)

Witness idea: 2-adic valuations as a structural fingerprint. Conjecture C7.3: the Mahler-spectrum valuation v_p^{max} separates "optimal" codes (28) from random codes (≈ 4).

v_p^{max} by code family — only Golay-type self-dual codes separate

Random-ensemble baseline (100 trials, $n=16$, $k=4$): mean 4.0, range 2–10 · Source: ACRP-06, 10.5281/z



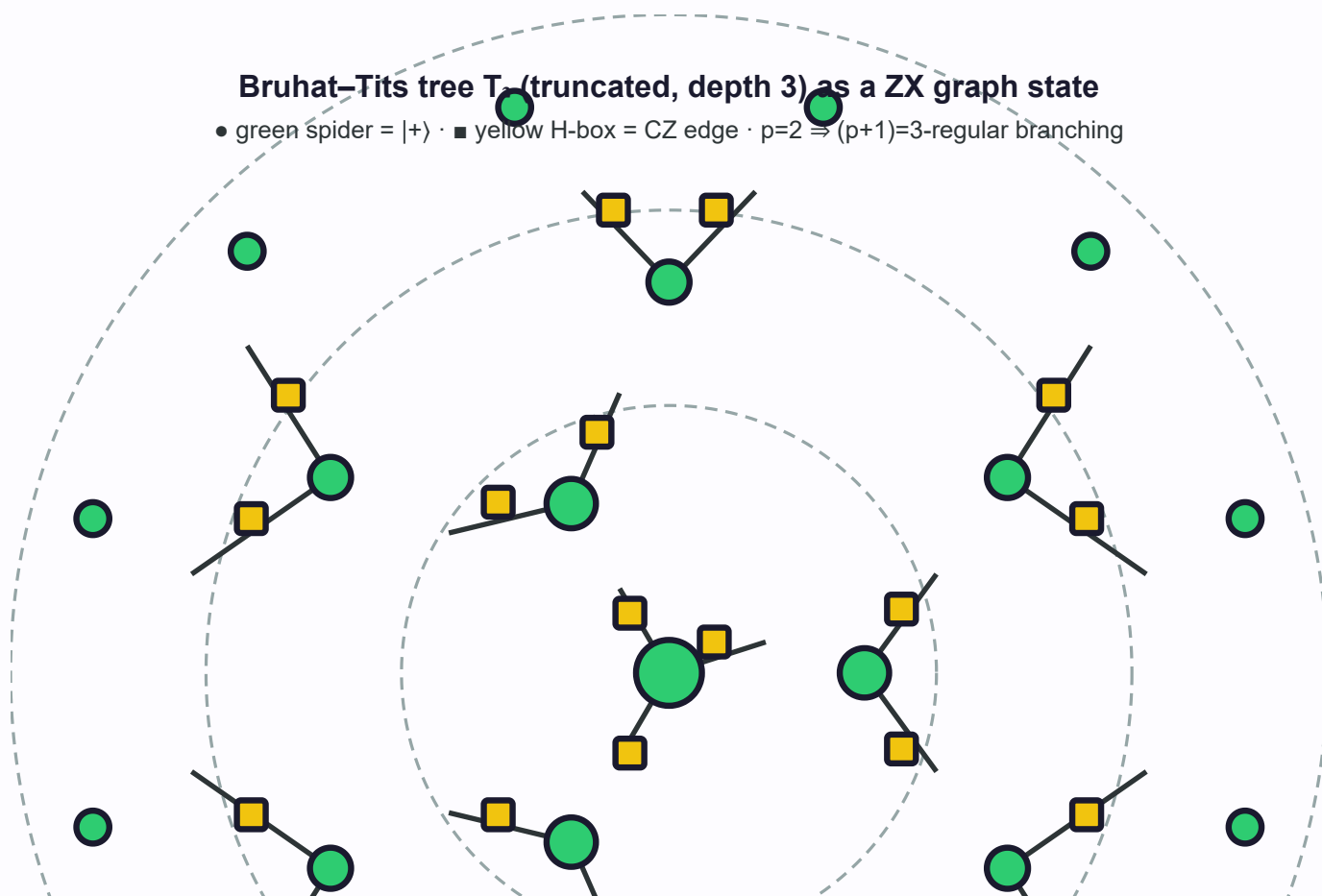
- Spiders
- Pauli w
- Gadgets

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Bruhat–Tits tree T_2 (truncated, depth 3) as a ZX graph state

• green spider = $|+\rangle$ · ■ yellow H-box = CZ edge · $p=2 \Rightarrow (p+1)=3$ -regular branching



References (Zenodo concept DOIs — verified 2026-08-16)

Ultrametric Code Spaces — 10.5281/zenodo.21824194
 Archimedean Shadows (v1.11) — 10.5281/zenodo.21809888
 Ultrametric Foundations (C7.3) — 10.5281/zenodo.21193003
 ACRP-06 v_p^{max} extension — 10.5281/zenodo.21737221
 Adelic Cross-Domain Program v5 — 10.5281/zenodo.21691414
 "What Remains" — 10.5281/zenodo.21922812

ZX references (arXiv — verified 2026-08-18)

Wan–Price–Yao, Holographic codes seen through ZX-calculus — arXiv:2601.04467
 Wan, Iteratively decoded magic state distillation — arXiv:2410.17992
 van de Wetering, ZX-calculus for the working quantum computer scientist — arXiv:2012.13966
 Duncan–Kissinger–Perdrix–van de Wetering, Graph-theoretic simplification — arXiv:1902.03178